

3.

$$C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$$

$$b = 4a$$

$$C_0 = \frac{2\pi\epsilon_0 l}{\ln(4)}$$

$$\begin{aligned} a &\rightarrow a/2 \\ b &= 8a \end{aligned}$$

$$C_1 = \frac{2\pi\epsilon_0 l_1}{\ln(8)}$$

$$\begin{aligned} \ln 8 &= \ln(4 \cdot 2) \\ &= \ln(4) + \ln(2) \end{aligned}$$

$$\frac{C_0}{C_1} = 1 \Rightarrow \frac{2\pi\epsilon_0 l / \ln(4)}{2\pi\epsilon_0 l_1 / \ln(8)} = 1$$

$$\frac{l}{\ln(4)} = \frac{l_1}{\ln(8)}$$

$$\Rightarrow l_1 = \frac{\ln(8)}{\ln(4)} l$$

$$= \left[\frac{\ln(4)}{\ln(4)} + \frac{\ln(2)}{\ln(4)} \right] l$$

$$= \left[1 + \frac{1}{2} \right] l \Rightarrow l_1 = 1,5 l$$